AlphaRise: A compassionate neurofeedback game for MS fatigue management

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Background

Multiple sclerosis (MS) related fatigue affects 75-90% of people with MS and is the most debilitating symptom. Current strategies show modest effectiveness. AlphaRise introduces adaptive, engaging, and compassionate neurofeedback — designed to address fatigue through real-time brain-game interaction.

Research Gap:

Current neurofeedback systems rely on fixed thresholds with minimal gamified engagement and no fatigue-aware response.

Survey (N = 146 pwMS)

MFIS -- 146 participants, 21 questions, 3 domains | alpha = 0.955

Physical fatigue
Weak muscles, needing more rest — 24.78 / 36 — 68.8%

Cognitive fatigue
Slowed thinking, difficulty concentrating — 22.60 / 40 — 56.5%

Psychosocial fatigue
Limited social activities — 4.99 / 8 — 62.4%

12% strategies very effective

96% willing to try games

R^2=.01 predictors non-significant

52.4 mean MFIS fatigue

Key Finding:

Behavioral strategies cannot predict or control MS fatigue (R^2=.01). AlphaRise engages neural mechanisms directly.

How the survey shaped AlphaRise:

- 73% want "easy to use" -- minimal UI
- 65% prefer "calming visuals" -- soft palette
- 80% prefer sessions under 10 min
- 74% already play video games

MS-informed design choices

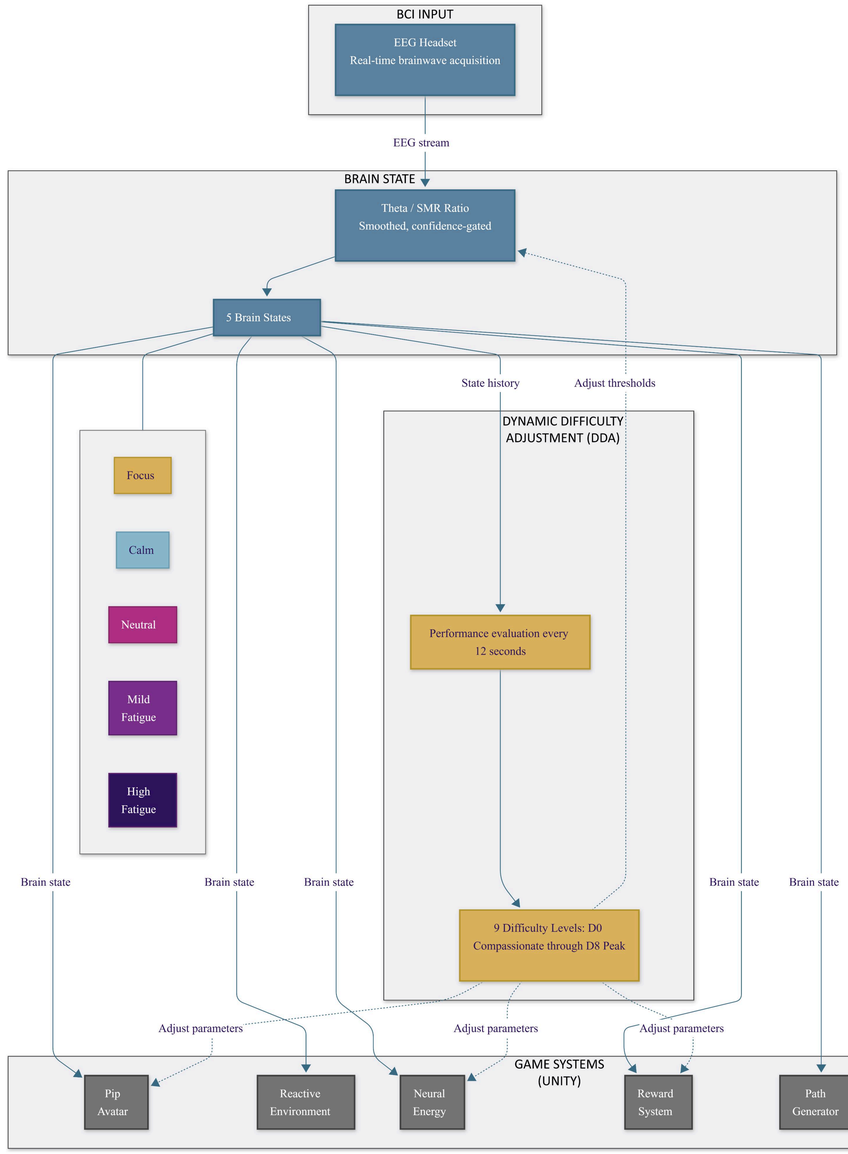
Accessible color palette
No red-green -- MS optic nerve safe

Blue-yellow dominant
High contrast for visibility
Desaturated UI, saturated game

- 5-min sessions (vs. 15-30 min prior)
- Calming 2d art, low cognitive load

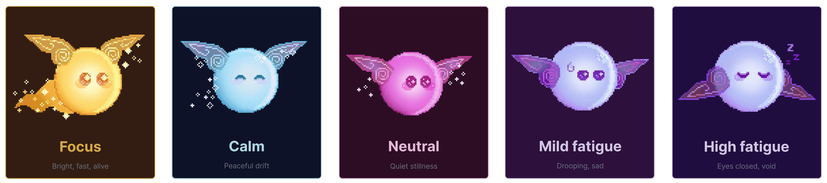
Closed-loop neurofeedback system

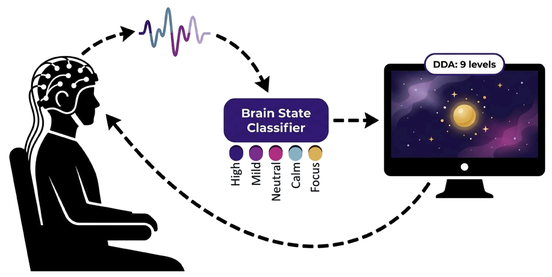
Neural activity classified into 5 states, adjusting difficulty every 12 seconds



Pip -- 5 brain states

Avatar externalizes the invisible experience of fatigue

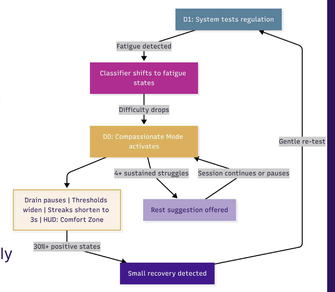




"It is okay to rest. Your brain is telling you something."

Compassionate Mode

When a player cannot sustain regulation, the game enters Compassionate Mode (D0) -- supporting rather than punishing. Energy drain pauses, classification thresholds widen, and streak requirements drop to 3 seconds, allowing even small moments of regulation to register as progress. The HUD displays "Comfort Zone" rather than any failure indicator.

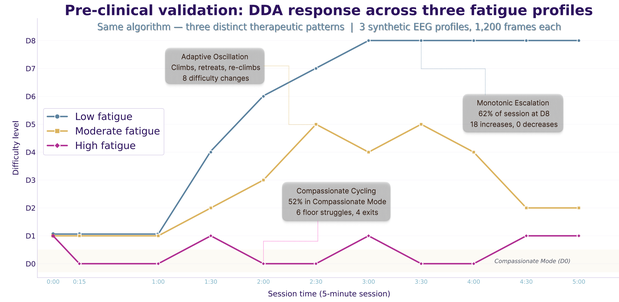


The system continuously re-tests: when 30% positive states are detected, it returns to D1. If fatigue reasserts, it drops back to D0 immediately -- creating a gentle challenge-support rhythm that respects the energy conservation needs of MS patients.

In pre-clinical validation, the high fatigue profile spent 52% of the session in Compassionate Mode.

Pre-clinical validation: DDA response across three fatigue profiles

Same algorithm — three distinct therapeutic patterns | 3 synthetic EEG profiles, 1,200 frames each



Comparison with prior systems

System	Adaptation	Lvls	Fatigue resp.
Birch Axon (2022)	Fixed	1	None
StoPain (2024)	Fixed	1	None
Kober MS (2019)	Clinician	Man.	None
AlphaRise (2026)	**Auto/12s**	**9**	**Compassionate Mode**

Future directions

- Clinical pilot trial with pwMS
- Live BCI integration pipeline
- Pre/post MFIS comparison
- Scalable home-based deployment

References
Scan for full reference list and acknowledgments.

vMed26
AMXRA Poster Session

Games that care, for stories that matter.

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